Suche

Archive

Past issues

Past articles

- Digital Acceptance and Commitment Therapy for the Treatment...
- Digital Acceptance and Commitment Therapy for Chronic Pain
- Specialties |
- For authors |

- References
- Metrics
- Citations

In Germany, approximately 800 000 elderly people (1) live in around 13 600 nursing homes; this population faces significant shortcomings in terms of ophthalmic care, already today and in the future (2). The OVIS and SÄVIP studies showed that a large number of elderly people have eye disorders requiring treatment (cataract, glaucoma, and age-related macular degeneration [AMD]), but only very few of these visit an ophthalmologist (3, 4), primarily due to limited mobility and a lack of assistance. The TOVIS (*TeleOphthalmologische Versorgung In Seniorenheimen* [tele-ophthalmology care in retirement homes]) study investigates a tele-ophthalmological model in which trained optometrists, among others, carry out multimodal imaging in care facilities and transmit the results to ophthalmologists for tele-diagnsosis in order to overcome the hurdles faced by the elderly in terms of presenting to and receiving treatment in ophthalmology practices and departments.

Methods

After giving their written informed consent, elderly people living in three nursing homes in Bonn were assessed in March 2023 by means of questionnaires (recording, for example, visual complaints, general/eye diseases, medication, care level, mobility, falls, visits to an ophthalmologist) and standardized ophthalmological tests (autorefraction, visual acuity, intraocular pressure measurement, anterior segment/fundus photography, optical coherence tomography [OCT], among others). All results were transmitted in pseudonymized form to the Department of Ophthalmology at the University Hospital Bonn (Harmony telemedicine platform), where the image data were checked for image quality (scale of 1—5; two assessors) and evaluated for ophthalmological findings. Results and diagnoses were sent to participants by post together with a recommended course of action (eyeglass lenses to be adjusted, urgent presentation to an ophthalmologist, etc.). Follow-up treatment was provided by ophthalmologists in private practice or the hospital eye department. PerplexityAI and ChatGPT were used to write the manuscript, but all contents were carefully verified.

Results

Out of all residents, 109 underwent the check-ups *(Figure)*. Visual acuity could be determined in 89.9% of eyes (n = 218) and intraocular pressure in 100% of eyes. The anterior segment could be assessed in 92.7% and fundus

Discussion

In addition to highlighting potential advantages of tele-ophthalmological care of elderly individuals living in residential facilities, the TOVIS study also showed the viability and effectiveness of the model (89—92.7% of images could be evaluated). Only the limited mobility of some participants (34%) and high level of care (39.4%), typical for the residential structure in facilities of this kind, had a negative effect on the quality of examinations (5). In the future, flexible devices (for example, hand-held slit lamps, portable OCTs) could be used to enable everyone to receive adequate examinations. As a first step, on-site check-ups can be used to assess urgency and determine the therapeutic intervention required; a specific recommendation can then be made to refer the patient to a practice or specialist department, thereby relieving the burden on practices, care facilities, and those affected. What was striking was that a large proportion of those examined were not sufficiently informed about their eye disorders and the need for treatment, which, in the case of progressive disorders (such as AMD and glaucoma), can lead to severe visual impairment and blindness (for example, in exudative AMD: > 70% within 3 years without treatment). Limited mobility and the subjectively perceived lack of contact with family/friends (OR 2.0 [0.65; 6.25]) are factors that hinder visits to the ophthalmologist. TOVIS can remove these hurdles and is integrated in a prospective multi-stage research project (InnOCaRe—INNovative Ophthalmologic Care in

Retirement homes), which is also intended to facilitate contact with practices/departments in the event that treatment is needed and overcome the obstacles posed by mobility. As part of this, the cognition of study participants will also be assessed and an economic evaluation of the new care model carried out. In the long term, this could result in a considerable bundling of resources and provide relief to partners and family members involved—with the elderly individuals themselves being the main winners in terms of receiving adequate ophthalmological care that will help them maintain their visual performance even at higher age.

Leon von der Emde*, Claus Gruber*, Lilith Arend, Robert P. Finger, Jennifer Nadal, Matthias Schmid, Frank G. Holz, Thomas Ach

*These authors share first authorship.

Augenklinik, Universitätsklinikum Bonn, Germany (von der Emde, Arend, Holz, Ach) thomas.ach@ukbonn.de;

Mirantus Health GmbH, Berlin, Germany (Gruber);

Augenklinik, Universitätsklinikum Mannheim, Germany (Finger);

Figure
Study enrollment process and comorbidities
Enlarge All figures

Table
Eye test results
Enlarge All figures

1. Blüher S, et al.: Bundesgesundheitsblatt Gesundheitsforschung Gesundheitsschutz 2023; 66: 490–7. CrossRef MEDLINE PubMed Central
2. Wolfram C, et al.: Dtsch Arztebl Int 2019; 116: 289–95.
3. Fang PP, et al.: Ophthalmologe 2017; 114: 818–27. CrossRef MEDLINE
4. Hallauer J, et al.: Vincentz Network Marketing Service 2005; 1–64.
5. Morse AR: Ophthalmology 2014; 121: 809–11. CrossRef MEDLINE

Info

- PDF-Version
- ePaper
- Table of contents
- German version
- Request permissions

Specialities

- Geriatrics

- Ärztestellen
- CME

- Jobs
- Anzeigenaufgabe
- Mediadaten
- Abonnement

- Impressum
- Kontakt
- Datenschutz & Datensicherheit
- Cookies
- AGB

Sie finden uns auch auf:

RSS-Feed abonnieren: